LESSON 11 · MISSION GUIDE

The Vault Beneath

SQL: ask the database a question, get rows back. The language every data team speaks.

From **Orrin of the Clockworks**
To **Jen, Keeper of the Guild Ledger**
By **raven, from the bottom of the stairs**

Keeper, under the guild hall is the Vault. Not gold: records. Every hero, every quest, every party, every region, in one file called `guild.db`. It's a **database**. You don't open it like a workbook; you ask it questions in **SQL**, and it answers in rows. Dalvric used four hundred lines. Please use fewer.

SQL is older than me and will outlive us both. Every guild, bank and shipping house in the realm has a Vault like this, and the people who can talk to it never go hungry. Python has a key to it built in. Mind the third step. — O.

What you'll be able to do after today

- [ ] Open a SQLite database from Python and run a query: `con.execute(sql).fetchall()`
- [ ] **Pick columns** with `SELECT` and **filter rows** with `WHERE`: `= <> < > AND OR NOT`
- [ ] Filter smarter: `IN`, `BETWEEN`, `LIKE`, `IS NULL`
- [ ] **Sort** and **cut**: `ORDER BY` (ASC/DESC, several columns), `LIMIT`, `DISTINCT`
- [ ] **Summarise**: `COUNT SUM AVG MIN MAX ROUND`
- [ ] Build a **PivotTable** in SQL: `GROUP BY` and `HAVING`

Why this matters

- **Real data lives in databases.** Every analytics job ad asks for SQL; it's the most useful single skill after Excel.
- **You already know the ideas.** WHERE is AutoFilter, GROUP BY is the PivotTable. Today is new words for old friends.

Four big ideas

1. A query is a sentence	`SELECT name, level FROM heroes WHERE class = 'Ranger'`: which columns, from which table, which rows.
2. The clause order is fixed	SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, LIMIT. Always in that order.
3. WHERE before, HAVING after	`WHERE` filters rows before grouping; `HAVING` filters the groups (it can use COUNT, AVG ...).
4. Python asks, SQL answers	`con.execute(sql)` runs it; `.fetchall()` is a list of tuples; you print with f-strings.

Three things that will bite you

- Text in **single quotes**, matched exactly: `'Success'` is not `'success'`. No quotes: SQL thinks it's a column.
- Whole number ÷ whole number drops the decimals in SQLite. Use `AVG()` or `* 1.0`.
- `AND` binds tighter than `OR`. Bracket your ORs.

How today works

15 MIN
Lecture
Lars opens the Vault and asks it the War Table questions, in SQL

60 MIN
Exercise sheet
Seven ravens, seven queries (or twenty)

15 MIN
Pages from the Grimoire, page 11
"vault_report.py", from inside a chest lid

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Orrin	Knock on the Vault	`connect`, `execute`, `fetchall`, `COUNT(*)`
2	Siv	The AutoFilter	`SELECT WHERE AND OR <> ORDER BY`
3	Pell	Sharper filters	`IN BETWEEN LIKE IS NULL NOT`
4	The Queen	Top of the board	`ORDER BY … DESC`, `LIMIT`, `DISTINCT`
5	Pell	Counting the hoard	`COUNT SUM AVG MIN MAX ROUND`
6	Siv	The PivotTable, in SQL	`GROUP BY`, `HAVING`
7 ⭐	Siv	Worth it?	aliases, `* 1.0`, `HAVING` + `LIMIT 1`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if Siv still wants her yes or no), each printing the expected output
- [ ] Dalvric's vault_report runs with real wins and real pennies, one commit per bug
- [ ] Everything is pushed to `lesson-11` and you've opened a pull request

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RUNE
Rune of the Vault

Next lesson: The Deep Archive. Today each question used one table. Next time you join them: heroes to parties to quests to regions, with one region missing from the map. Plus questions inside questions, and the Vault straight into pandas.